

1 Markov Decision Processes

A Markov Decision Process (MDP) is a mathematical framework used to model decision-making in situations where outcomes are partly random and partly under the control of a decision maker. An MDP is defined by the tuple (S, A, P, R, γ) , where:

- S is a finite set of states.
- A is a finite set of actions.
- $P(s'|s, a)$ is the state transition probability function, which gives the probability of transitioning to state s' from state s after taking action a .
- $R(s, a)$ is the reward function, which gives the immediate reward received after taking action a in state s .
- $\gamma \in [0, 1)$ is the discount factor, which determines the importance of future rewards.

A policy $\pi(a|s)$ defines the probability of taking action a when in state s .

The return G_t is defined as the total discounted reward from time step t onward:

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \quad (1)$$

The goal of solving an MDP is to find a policy $\pi : S \rightarrow A$ that maximizes the expected cumulative return from each state.

Note: All of the above assume a discrete state and action space, as well as discrete time steps.

2 Bellman Equations

The value function $V_\pi(s)$ for a given policy π is defined as the expected return when starting in state s and following policy π :

$$V_\pi(s) = \mathbb{E}_\pi[G_t | S_t = s] \quad (2)$$

The action-value function $Q_\pi(s, a)$ is defined as the expected return when starting in state s , taking action a , and thereafter following policy π :

$$Q_\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a] \quad (3)$$

The action-value function is often referred to as the Q-function.

Bellman equations provide a recursive relationship for the value functions. The Bellman equation for the value function $V_\pi(s)$ is given by:

$$V_\pi(s) = \sum_a \pi(a|s) \sum_{s', r} P(s', r | s, a) [r + \gamma V_\pi(s')] \quad (4)$$

The derivation of the Bellman equation for the value function is as follows:

$$\begin{aligned}
V_\pi(s) &= \mathbb{E}_\pi[G_t | S_t = s] \\
&= \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} | S_t = s] \\
&= \sum_a \pi(a|s) \sum_{s', r} P(s', r | s, a) [r + \gamma V_\pi(s')]
\end{aligned}$$

3 Policy Iteration

Policy iteration is an algorithm used to find the optimal policy for a Markov Decision Process (MDP). It consists of two main steps: policy evaluation and policy improvement. The algorithm iteratively evaluates the current policy and then improves it until convergence to the optimal policy.

Algorithm 1 Policy Iteration

- 1: Initialize policy π arbitrarily
- 2: **repeat**
- 3: **Policy Evaluation:** Compute $V_\pi(s)$ for all states s using the Bellman equation

$$V_\pi(s) = \sum_a \pi(a|s) \sum_{s', r} P(s', r | s, a) [r + \gamma V_\pi(s')] \quad (5)$$

- 4: **Policy Improvement:** Update policy π using the greedy policy derived from the value function

$$\pi'(a|s) = \begin{cases} 1 & \text{if } a = \arg \max_a Q_\pi(s, a) \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

- 5: **until** Convergence
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